



AI for Scientific Integrity: Detecting Paper-Mill Publications Through NLP and Scholarly Metadata

Alessandra Vellucci Solari, Emil Shahbazov, Satyaki Sikdar, Daniel Moreira

INTRODUCTION

The rapid growth of scientific publications has been accompanied by an increase in fraudulent research practices, particularly the emergence of **paper mills** — organizations that produce and sell fabricated manuscripts. These papers often mimic legitimate research in structure, language, and citation behavior, making manual detection increasingly difficult. As a result, paper-mill publications threaten the credibility of scientific literature and undermine trust in academic research.

Research Gap: Despite growing awareness, existing detection methods rely heavily on manual investigation or post-publication retractions. Current automated approaches are limited in their ability to capture both **linguistic patterns** and **structural metadata signals** associated with paper-mill activity. There is a lack of scalable, AI-driven solutions that integrate textual analysis with scholarly metadata to proactively identify suspicious publications.

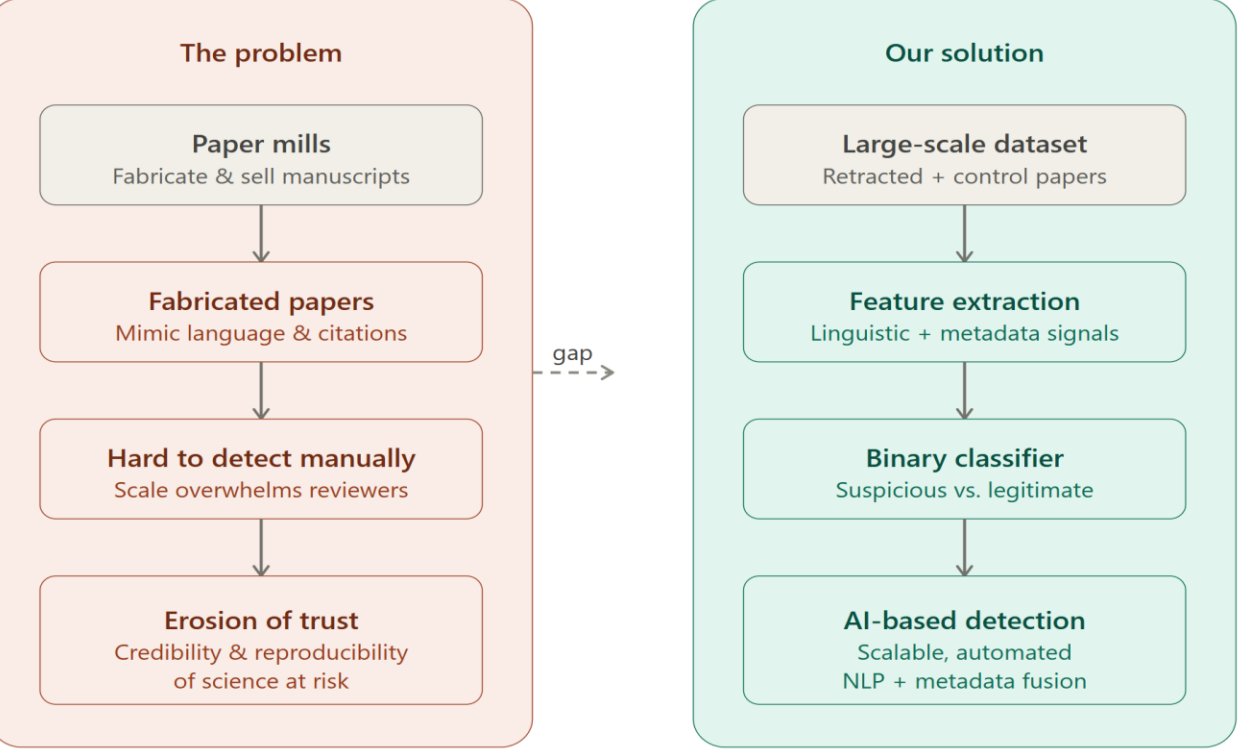
Our Contribution: We construct a **large-scale dataset** of scientific publications for studying paper-mill activity. We curate:

- Retracted papers associated with potential paper-mill indicators
- A large control group of non-retracted papers from OpenAlex

We define a **binary classification framework** (suspicious vs. legitimate papers) to support future modeling. We perform **initial exploratory data analysis (EDA)** to identify distinguishing patterns in:

- Authorship structure
- Publication trends
- Metadata distributions

We lay the **foundation for future AI-based detection models**, integrating NLP and metadata features



RESULTS

Dataset Statistics

- Paper-mill-related retracted papers: ~7,000+
- Non-retracted papers:
 - ✓ 1× dataset: ~7,000
 - ✓ 10× dataset: ~70,000
 - ✓ 100× dataset: ~700,000.

Preliminary Findings

- Suspicious papers exhibit:
 - ✓ Distinct **author count distributions**
 - ✓ Concentration in specific **publication periods**
 - ✓ Patterns suggesting **structured production behavior**
- Non-retracted papers show:
 - ✓ Greater variability in authorship and metadata
 - ✓ More diverse publication patterns

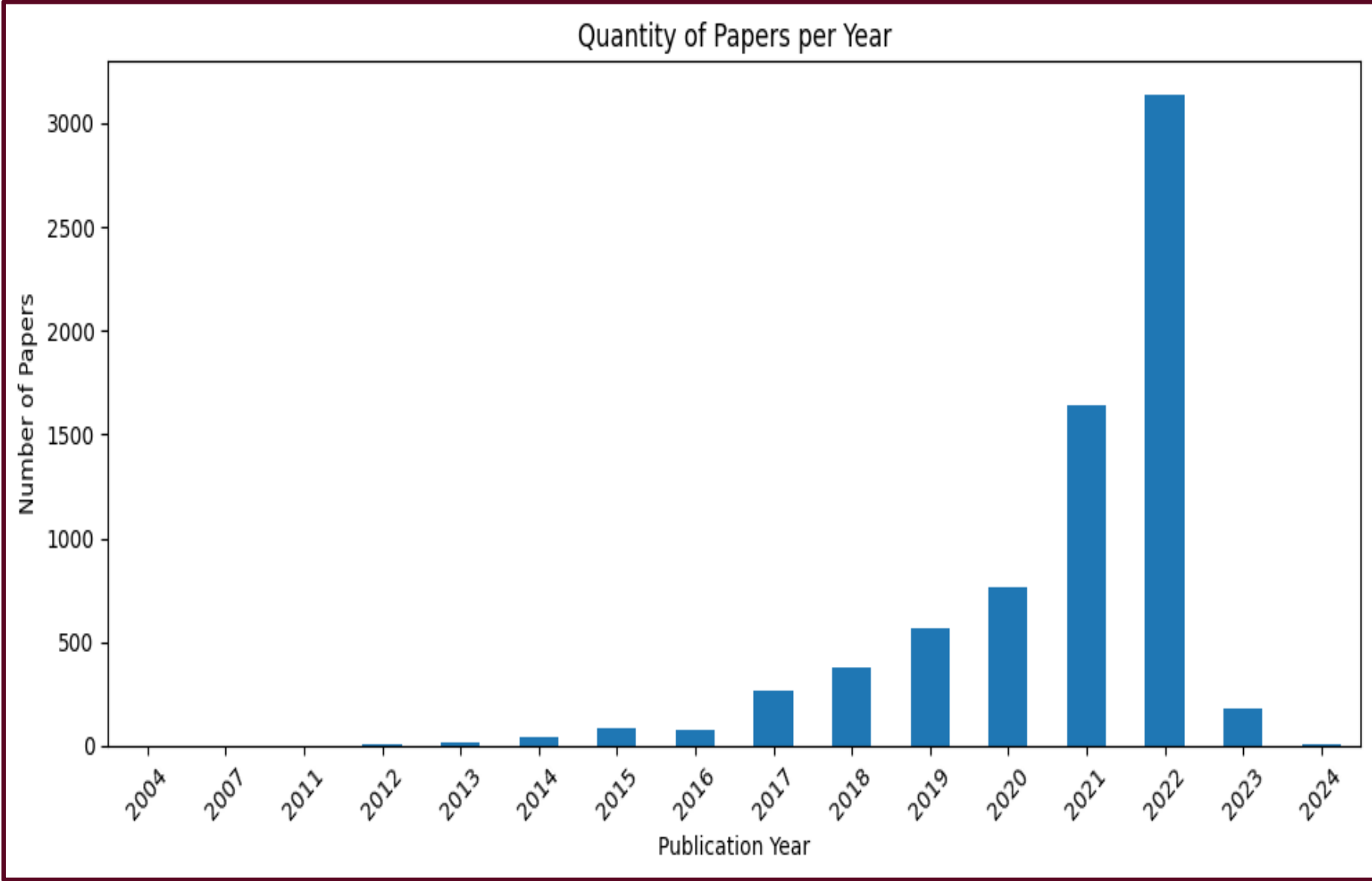


Figure 1: Retracted papers with retraction nature per Year from 2004 - 2024

Key Insight

These differences suggest that metadata signals alone may provide useful indicators for detecting potential paper-mill activity, motivating further AI-based modeling.

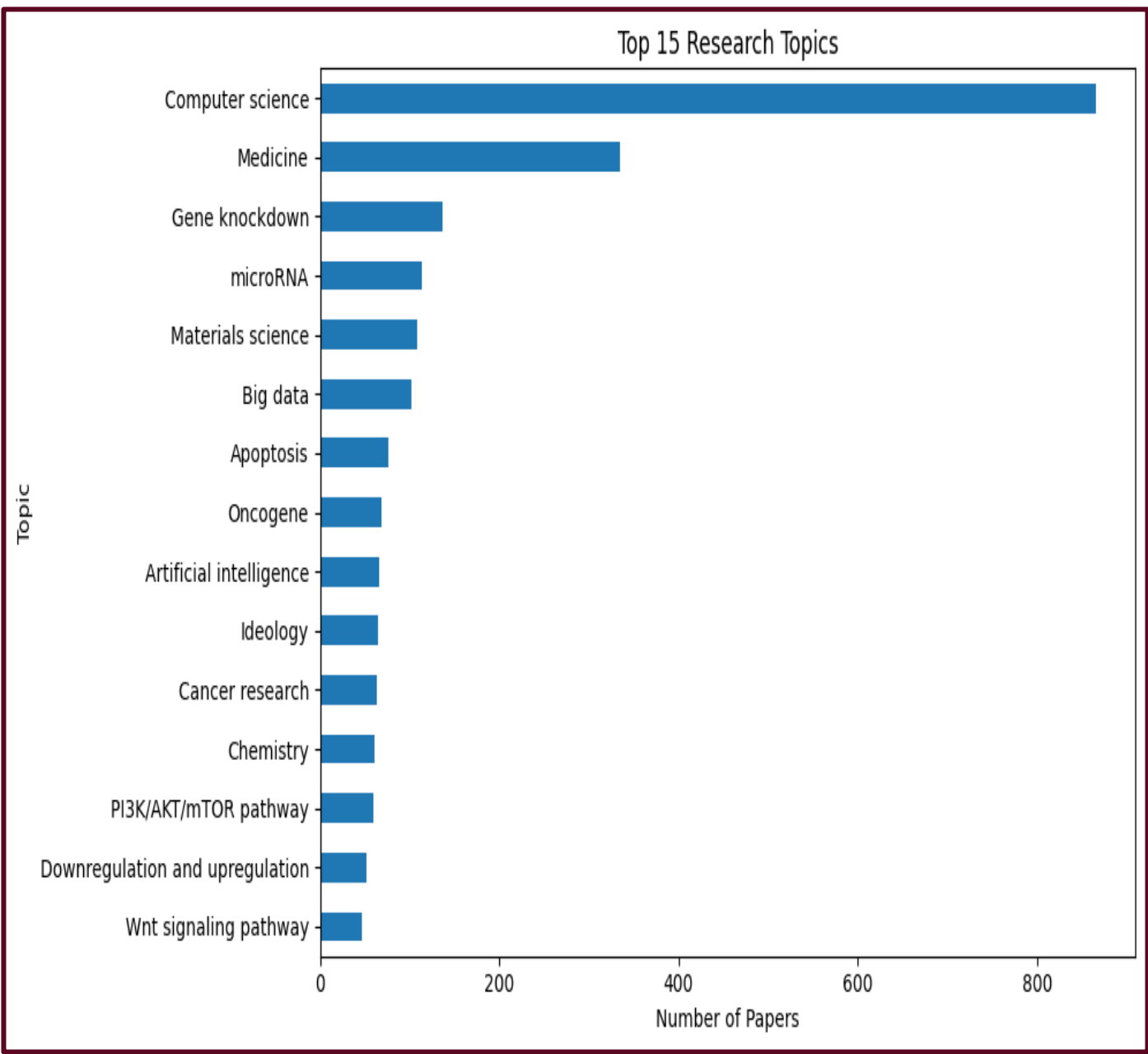


Figure 2: Top 15 Topics of Retracted Papers with retraction nature

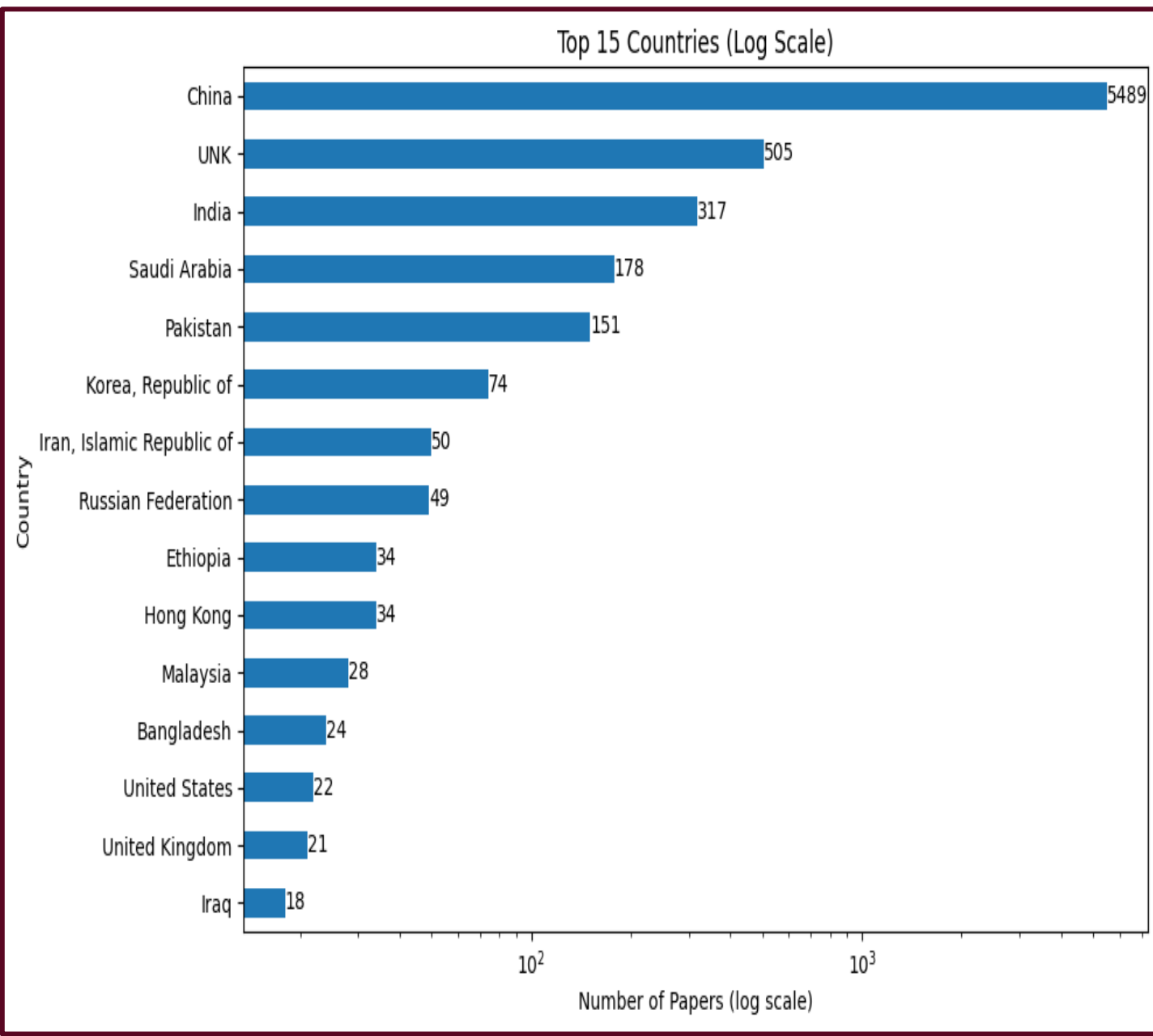


Figure 3: Top 15 Countries with Retracted Papers

METHOD

Data Collection

Suspicious papers (positive class):

- Extracted from retraction databases (e.g., Retraction Watch)
- Filtered for paper-mill-related reasons

Legitimate papers (negative class):

- Collected using the OpenAlex snapshot API
- Filtered by:
 - ✓ Non-retracted status
 - ✓ Publication years (2004–2024)
 - ✓ Comparable domains

Dataset Construction

- Cleaned and standardized DOIs
- Matched and validated records across datasets
- Built multiple dataset scales:
 - ✓ 1× (balanced)
 - ✓ 10×
 - ✓ 100×

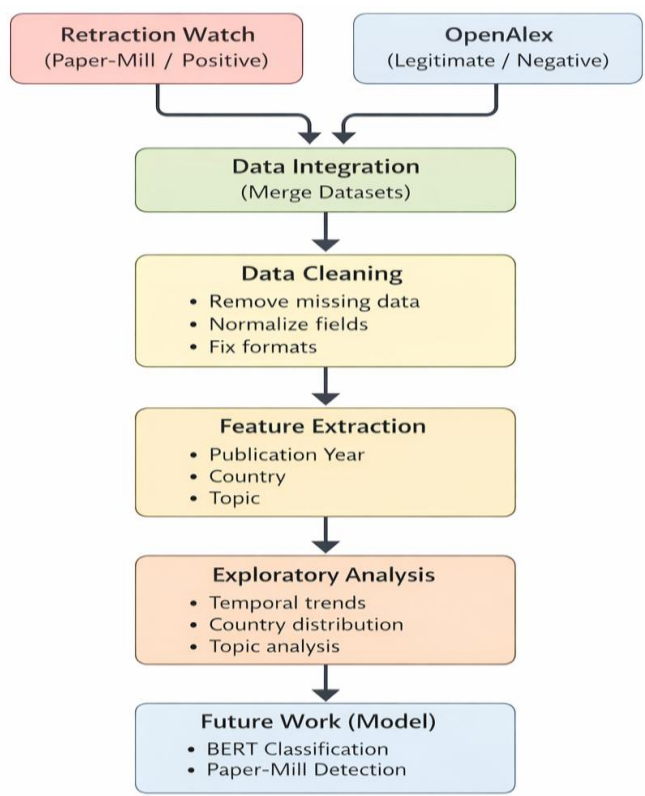
Feature Preparation

- Extracted:
 - ✓ Titles and abstracts (for future NLP use)
 - ✓ Metadata features such as:
 - Number of authors
 - Publication year
 - Venue information

Exploratory Data Analysis (EDA)

We analyze differences between retracted and non-retracted papers through:

- Distribution of number of authors
- Temporal publication trends
- Metadata variability



DISCUSSION

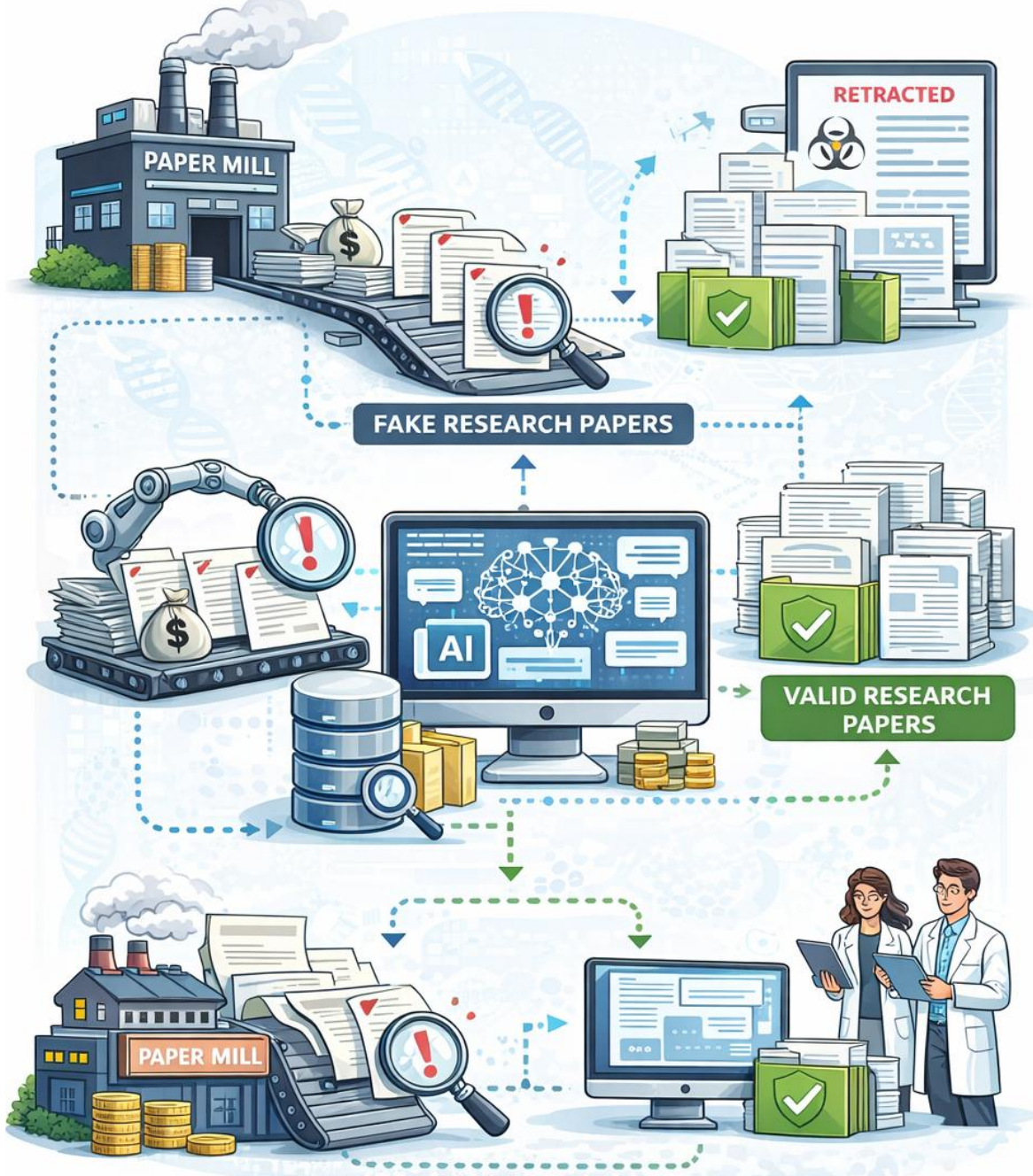
This work focuses on building the data foundation necessary for AI-driven detection of paper-mill publications. Our preliminary analysis highlights measurable differences between suspicious and legitimate papers, particularly in metadata patterns, suggesting that these features can play a key role in automated detection systems.

Limitations

- Reliance on retracted papers as proxies for paper-mill activity
- Not all paper-mill papers are identified or retracted
- Potential bias in dataset construction
- Current work does not yet include trained predictive models

Future Work

- Develop and fine-tune transformer-based NLP models (e.g., BERT) for text classification
- Integrate textual and metadata features into a unified model
- Predict paper-mill likelihood prior to retraction
- Expand dataset with additional sources and full-text data
- Explore network-based features (citations, co-authorship graphs)



ACKNOWLEDGEMENTS

- We extend our gratitude to Abigail Daman, Christina Truszkowski, and Joao Cardenuto for their collaboration in the research.

REFERENCES

- Crossref. (2024). "Retraction Watch data (daily updates)". GitLab. <https://gitlab.com/crossref/retraction-watch-data>
- Priem, J., Piwowar, H., & Orr, R. (2022). "OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts". ArXiv. <https://arxiv.org/abs/2205.01833>
- Sikdar, S. (2026). "OpenAlex Preprocessing". GitHub. <https://github.com/satyakisikdar/openalex>